

Complaint Tracker

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

The **Complaint Tracker** is a text-based system designed to efficiently manage and search large collections of customer complaints. The system organizes complaints using unique complaint IDs, categories, descriptions, statuses, priorities, and resolutions. A structured corpus containing multiple complaint records is used to demonstrate efficient pattern searching. The project implements string-matching algorithms such as **Knuth-Morris-Pratt (KMP), Z-Algorithm, and Rabin-Karp with rolling hash** to locate specific complaint patterns within the corpus. These algorithms reduce unnecessary comparisons and improve searching efficiency compared with simple brute-force searching, especially when processing large text datasets. Each complaint is maintained as an individual file and all records are combined into a single corpus for large-scale testing. The project also provides search results that identify matching complaint records and their IDs. Overall, the Complaint Tracker demonstrates the practical application of linear-time and efficient string-search techniques for organizing, retrieving, and analyzing complaint data.

Signature of the Guide